

Experiment No. 25

Title

Prolog Program to Implement Monkey Banana Problem

Aim

To write a Prolog program to solve the **Monkey Banana Problem** using logical reasoning.

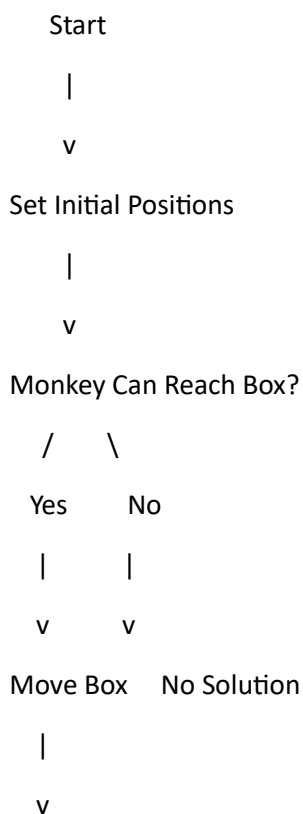
Objective

To understand how Prolog rules can be used to represent actions and solve a simple planning problem.

Algorithm

1. Start.
2. Define the initial positions of the monkey, box, and banana.
3. Check whether the monkey can reach the box.
4. Move the box below the banana.
5. Climb the box and get the banana.
6. Display the solution.
7. Stop.

Flowchart



Climb Box

|

v

Get Banana

|

v

Stop

Prolog Program

can_get_banana :-

at(monkey, floor),

at(box, floor),

at(banana, ceiling),

near(monkey, box),

write('Monkey moves the box below the banana'), nl,

write('Monkey climbs the box'), nl,

write('Monkey gets the banana'), nl.

Query

?- can_get_banana.

Sample Output

Monkey moves the box below the banana

Monkey climbs the box

Monkey gets the banana

true.

Result

The Prolog program successfully implemented the **Monkey Banana Problem** and produced the required sequence of actions.

Conclusion

Thus, the Monkey Banana Problem was successfully solved using Prolog rules and logical reasoning.